

At point D: object just reached the ground for the second time

4

C

$$s = ut + \frac{1}{2}at^2$$

$u = 0$ since the stone falls from rest.

$$2 = \frac{1}{2}at^2$$

$$\frac{4}{a} = t^2$$

$$t = \frac{2}{\sqrt{a}} \text{ ----- (1) where } t \text{ is the time taken for stone to fall on Earth}$$

Acceleration of free fall of moon = $\frac{a}{6}$, where a is the acceleration of free fall on Earth.

Sub $\frac{a}{6}$ into equation (1)

$$t_{\text{moon}} = \frac{2}{\sqrt{\frac{a}{6}}} = \frac{2}{\sqrt{a}}(\sqrt{6}) = t\sqrt{6}$$

5

B

Area under the force-time graph = change in momentum = impulse